What Does It Mean To Remember? Understanding Users' Perceptions and Practices with Long-Term Memory in AI Agents

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Abstract

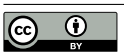
AI agents promise long-term personalized assistance, contingent on effective long-term memory (LTM). Yet little is known about users' perceptions, practices, and challenges with agent LTM. We interviewed LLM users, finding that interactions with LTM primarily involve attempts to influence and understand how the agent interprets information from memory, rather than managing the storage of discrete data. To test the alignment between these perceptions and current memory-management methods, we conducted a "matching game" user study, in which participants were asked to identify which specific memories the agent used to produce an output. Overall, participants failed to successfully match discrete data to interpretations, suggesting a worrisome misalignment between common "discrete data" memory-management approaches and users' conceptualizations of LTM as data interpretation. We discuss implications for users' abilities to reliably anticipate how personal data shapes AI behavior, and propose design pathways that help users form clearer links between data and interpretation.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

human-AI interaction, human-agent interaction, memory, personalization, language models



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1 Introduction

People are increasingly interacting with AI agents (e.g., Claude Cowork and OpenClaw) over longer periods of time—e.g., using them as personal AI assistants (e.g., [3, 29]) or 'coaches' for wellness [9, 10, 12, 14, 17, 35], exercise [22, 24, 26], life goals [21, 44], and work productivity [33, 37]. Users decide what to disclose, how to compartmentalize tasks, and how to recover from errors based on their working assumptions about what the agent will remember. The agent's ability to remember and work in collaborative 'complementarity' [40] with users over time requires the agent's *long-term memory* (LTM) [20, 36, 53]. However, the technical implementations of agent LTM and interactive memory controls may not match users' mental models. To align LTM mechanisms with users' mental models, we must (in line with [40]) understand users' perceptions and practices. Although the development of memory mechanisms in LLM-based agents has been extensively covered by prior research (e.g., [13, 48]), there is still little understanding of users' perceptions, practices, and challenges with agent LTM. Understanding users' perceptions and practices is therefore important for aligning the technical implementations of agent LTM with users' mental models, and for designing LTM mechanisms that preserve users' goals.

In the first study, we conducted interviews (N=21) with participants who regularly use personalized AI agents with LTM, and thematically coded these interviews to explore users' perceptions and mental models of these systems' memory mechanisms and

their practices for managing memory in these systems. Our findings shed light on the flow of information in an agent's LTM as a three-stage process consisting of (1) information *intake*, (2) *storage and management*, and (3) *retrieval and interpretation* (Figure 1). Our participants' reflections reveal that users' perceptions and mental models of agent memory are primarily influenced by Stage 3 (*retrieval and interpretation*), and as a result, users' practices with managing and exploring agent memory are largely an attempt to influence and understand how the agent retrieves and interprets information from its LTM. Participants described managing and organizing information into separate chat threads, defining custom agents for different tasks, repeating or emphasizing certain information in conversation, asking the agent to summarize what it knows, and several other practices. However, these practices have their pain points, and participants expressed their desire for more control and understanding of the retrieval and interpretation of information from LTM.

But do current systems allow users to understand the interpretation of information from LTM? Most current technical implementations of agent LTM and interactive memory controls focus on managing discrete memories (i.e., the user can see a list of stored facts that are inserted into user prompts). This implementation assumes that it will be obvious to the user how the agent interpreted those discrete memories, as the user can read the generated output and recognize cues, explanations, or references to the stored facts.

In the second study, we tested whether users can reliably map from discrete stored memory items to the agent's resulting interpretation in an output. We developed a system that allows users to include or exclude various memories from a list of 17–21 memories and generated LLM responses using only 3–5 memories. If discrete memory lists are sufficient to support traceability, users should be able to infer which memories were used to produce a given output when the search space is bounded and the interface provides direct control over inclusion and exclusion. Therefore, we conducted a within-subjects experiment (N=80) across various scenarios to determine whether users can identify the specific memories the agent used to generate the desired output. Our results show that although users made progress and correctly identified a few memories, they were unable to accurately identify the majority of them, meaning that they could not reliably connect discrete stored items to the agent's output-level interpretation. This indicates a concerning traceability/provenance gap in common "discrete data" memory-management approaches, and suggests that discrete-list controls may be structurally misaligned with users' Stage-3-oriented goals observed in Study 1.

We conclude by recommending that technological approaches to user interaction with agent LTM should focus at least as much on providing control and understanding of memory retrieval and interpretation as they do on memory intake, storage, and management. We also discuss implications for users' abilities to reliably anticipate how personal data will shape AI behavior, and propose design pathways that help users form clearer links between data and interpretation.

2 Related Work

While previous work has extensively covered memory in AI agents (see, for example, recent reviews by He et al. [13] and Zhang et al. [48]), it has predominantly focused on implementation and technical mechanics, i.e. *how* an agent can remember information relevant to a given user. For instance, particularly discussed strategies for maintaining an agent's memory include summarization (e.g., [39, 42]) and refinement (e.g., [52]) of conversational and interactional histories, maintaining vector databases and retrieving relevant conversations or information through semantic or linguistic matching (e.g., [16, 34, 53]), maintaining key-value pairs in traditional databases (e.g., [32]), or maintaining lists of 'facts' or 'observations' learned from user interactions (e.g., [2, 5, 27, 34, 43]). Beyond text chat, recent systems (e.g., [28]) also target users' personal, multimodal "captured memories" (e.g., photos, screenshots) and augment them with contextual links to support complex personal QA. However, despite the active development of research on the implementation of memory techniques, the questions of which memories and how they should surface in interactions with users remain unexplored. This gap becomes even more notable in the development of end-users' controls over agent memory [15].

Indeed, recent work on LLM companions also reports a desire for adjustable memory settings to control what is retained across sessions [46, 51]. Currently, the systems allow users to influence or modify agents' memories through, most often, direct manipulation of "remembered" data. HCI probes such as Memory Sandbox and Memolet expose remembered snippets as tangible UI objects that users can pin, rewrite, or delete [18, 45]. These designs recognize that memory is a joint resource, but early deployments provide little evidence of how people decide *what* to curate or how they judge the agent's subsequent use of curated memories. Some systems' memory mechanisms (e.g., [23]) are designed to mimic existing psychological models of human memory (e.g., [4]). Inspired by multi-store theories [4], some systems simulate episodic versus semantic stores, decay, and consolidation to achieve "human-like" forgetting curves [23]. Other systems (e.g., [32]) are designed to mimic existing technological metaphors, such as traditional operating-system memory. More recently, cognitive-AI frameworks explicitly organize long-term memory into modules for control, retrieval, and storage to improve context awareness (e.g., [41]). These metaphors justify algorithmic choices, yet they remain agnostic to whether end-users intuit or even value such cognitive fidelity—prompting a need to examine users' *mental models* of agent memory.

Across settings, remembering reliably boosts perceived quality of interaction, yet simultaneously heightens sensitivity to privacy and control. In a three-week field experiment, Cox et al. [8] showed that a chatbot that referred to prior conversations was judged more engaging and intelligent than one that appeared to be stateless. In a domain where continuity is especially valued, Jo et al. [20] found that a health chatbot's recalls fostered a sense of familiarity and trust, which in turn increased willingness to disclose.

The same mechanism, however, can feel intrusive depending on how and when prior material is resurfaced. Cox et al. [8] observed that verbatim callbacks—quoting users' exact words—often triggered privacy concerns even among participants who otherwise appreciated continuity. Compounding this, many users do not know

what information LLMs can retain or how it is used: Zhang et al. [47] report that most interviewees were unaware of cross-session memory and held misconceptions about its scope and visibility.

Finally, recent work shows that framing and timing modulate this trade-off. Cox et al. [7] found that presenting a bot as a persistent "familiar" versus an ephemeral "stranger" shifts disclosure comfort and enjoyment, with effects contingent on conversational order (e.g., starting with low-risk topics can make persistence feel beneficial rather than invasive). These studies suggest that perceived intelligence gains accrue when memory supports continuity and care, while privacy and comfort hinge on users' awareness, the form of the callback (verbatim vs. paraphrase), and the context in which it is reintroduced. These findings guide our focus on users' expectations for interpretation—not just what is stored, but which traces should re-enter the conversation and when.

Overall, while prior work on agents' LTM *engineers* sophisticated memory pipelines and *documents* initial user reactions, it has not yet explored *how people want agents to interpret and deploy stored information over time*, nor the ad-hoc tactics users adopt to influence that process. Achieving such an understanding could help designers build memory mechanisms for agents that involve users, consider their mental models (in line with [40]), and keep the human in the loop.

3 Study 1: Users Perceptions of AI Agents Memory Mechanisms

We conducted an interview study to understand users' perceptions and expectations regarding AI agents' memory mechanisms, as well as their practices for managing these systems' memories over longer periods of use.

3.1 Study 1 Method

We conducted semi-structured one-on-one interviews on Zoom (see Section A of the supplementary material for the interview guide) with 21 English-fluent participants (11 male, 10 female; mostly 21–35 years old) who use personalized AI agents with LTM on a regular basis. In this study, *agent LTM* referred to the agent's capacity to retain, retrieve, and use information from prior interactions across time and, where applicable, across conversations. The interview protocol covered participants' current uses of personalized AI tools, the role of memory in their AI use, their mental models of how agent memory works, their practices for influencing memory, and situations in which they wanted more awareness or control over memory. Interviews lasted between 30 and 90 minutes (50–60 minutes on average). All interviews were conducted by the same author. The guide in Section A of the supplementary material served as a flexible scaffold rather than a fixed script: the order and wording of questions varied with the flow of each conversation, and the interviewer asked unscripted follow-up questions to clarify or expand on participants' responses.

Participants were recruited through social media advertisements, email recruiting, postings in university and academic Slack channels, and snowball sampling. Section B of the supplementary material provides an overview of our participants. Most participants used AI tools at least once per day (18/21), while the remaining participants used them at least once per week (3/21), with use cases spanning productivity, research, coding, emotional support, companionship, language learning, planning, writing assistance, and information seeking. Each participant is identified as *'P#'* in the findings. Participants rated their own expertise with AI tools and their technical knowledge of LLMs on five-point scales; per-participant ratings, along with the specific tools each participant used, are listed in Section B of the supplementary material.

The interviews were recorded and transcribed for analysis. We conducted a thematic analysis of the data using an inductive coding approach [6]. Codes were developed to address the research question, then iteratively discussed, refined, and categorized into themes. For example, codes like *"user is surprised about how much the system remembers about them," "user wants the system's memories to be organized or disjoint by project, task, or domain,"* and *"user organizes memories by chat threads"* were created, representing users' perceptions, unsatisfied needs, and current practices with agent memory. During the categorization phase, we saw themes emerge around desires for system transparency, heterogeneity of users' memory needs, and user practices of organizing and separating the system's memories. The first and second authors coded the transcripts independently, and all authors then discussed, reviewed, and iterated on the codes together. Disagreements were resolved through discussion until consensus was reached. Because the coding was inductive and the codebook continued to evolve throughout this process, we did not compute an inter-rater reliability statistic.

3.2 Study 1 Results

We describe memory flow for an agent with LTM as a three-stage process (Figure 1). In the first stage, discrete data comes in as input from the user in the form of messages containing prompts, instructions, or other context. In the second stage, some of the discrete data that comes in is stored and managed by the agent into its LTM. This storage and management can happen in several ways, depending on the implementation of the agent or its memory infrastructure. Third, at some point in the future, the information that is saved to the agent's LTM is retrieved and interpreted in a new context (e.g., a new conversation, task, or interaction session). In our study results, participants described perceptions, mental models, practices, user needs, and challenges that apply to each of these three stages.

3.2.1 Users' Perceptions and Mental Models of Agent Memory. Regarding Stage 2 (*storage and management*) of the memory flow process (Figure 1), participants expressed that they tend to assume that the agent is either storing all information into memory or storing no information, depending on the agent in use and its current mode. Tools like ChatGPT and Claude save users' chat histories to their accounts, and these chat histories are accessible to the user for their own viewing, or even for continuing previous old chats if desired. This gives some users the impression that the system is at least capable of storing and managing all information that comes in as user input to LTM. In other cases, such as when the system does not allow users to access previous chat sessions (e.g., ChatGPT's "temporary chat" mode), some users assume that the system is not storing or managing any of the information it takes in for longer-term use. Other users speculate that the company providing the agent is collecting and storing information from chat sessions, even if those chat histories are no longer visible to the user

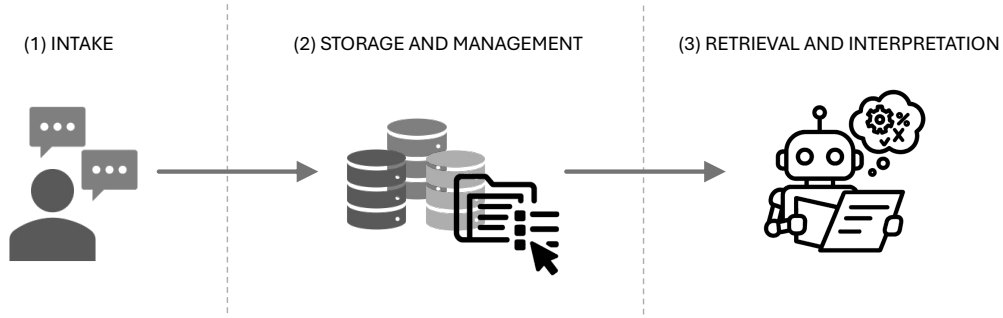


Figure 1: The process of memory flow for an agent with LTM. First, discrete data (in the form of messages or other context information) come in as input from the user (Stage 1: 'Intake'). Second, some of this discrete data is stored and managed into the agent's LTM (Stage 2: 'Storage and Management'). Third, at some point in the future, this information is retrieved from LTM and interpreted by the agent in a new context (Stage 3: 'Retrieval and Interpretation').

after the session ends, reflecting some users' general mistrust of the data-collection practices of technology companies. While users' perceptions and assumptions of how the agent operates in Stage 2 of the memory flow process are relatively clear and often based on what information from past interactions is visible to the user in the interface, users' perceptions of what the agent is doing in Stage 3 (*retrieval and interpretation*) are less clear. For instance, many of our participants expressed that they do not fully understand how, or even whether, the information the agent collects over time influences its outputs over the long term.

> *"It would be interesting, like I mentioned,* ***to know which parts of the memory it has about me go into, like, which responses****, and to understand more about how it decides which portions [...] of that list [of memories] is like... you know, [it is] 100 or 200 items... like, that's a lot of things to remember about me. And to decide, like, which parts of that list actually go into a response versus not."* — P3

When the agent does not provide full transparency on how it is retrieving, interpreting, and using information from its LTM, users tend to develop their own assumptions about how the agent is doing this. For instance, most of our participants (P1-P6, P8, P13, P15-P20) assumed that if the interface provided the ability to create multiple chat threads (as ChatGPT and Claude do), then the agent was able to retrieve memories from within the current chat thread better than information across chat threads.

Users' assumptions about agents' abilities to retrieve old memories were also largely based on the agents' outputs. For instance, several participants (P9, P12, P14, P18, P21) perceived that the agents they used adapted their conversational styles to users' preferences, or referred to previous knowledge or learned facts about the user, either explicitly or implicitly, during conversations. When this happened, participants assumed that the agent was more capable at retrieving and using old memories.

> *"****[It] will sometimes give you hints or clues that [it's] remembering something*** *from our previous conversations. For instance, if I'm working on a project, and I ask a question that's related to something we discussed earlier, it might respond with something like 'as you previously discussed' [...] So that phrase, 'as previously discussed',* ***is a great indication that it has drawn in our previous conversation and recalling specific details.****"* — P9

3.2.2 Users' Practices of Memory Management. Our findings reveal that users take a variety of approaches to managing memory in agents. For instance, because participants tended to assume that the agent would remember information from within the current chat session better than information from other chat sessions, several participants (P1-P6, P13, P15-P18, P20, P21) expressed that they manage conversations about different topics or tasks into different chat sessions, so that the agent would have easier access to memories that are relevant to each specific task. Some participants customized chat sessions by feeding the agent custom instructions (P1-P6, P18) or specific knowledge or terminology related to the task (P3, P5, P10, P13, P14, P18, P19) at the beginning of the session. Others (P1, P2, P4, P6, P15, P16) expressed that they would copy and paste system instructions or relevant context from previous chat sessions into new chat sessions when necessary, in order to selectively preserve some of the agent's desired behaviors or memories from previous sessions inside the new session.

While some participants were intentional in organizing different topics and tasks into multiple chat sessions, others (P8, P9, P14, P19) described using a single chat session for all of their communication with the agent, even if they were using the agent for different tasks or communicating about different topics. Commonly-cited reasons for doing this include to *"preserve [the] continuity"* (P8, P9) of the conversation or because the user thought that it would be beneficial for the agent to have access to all of its past memories no matter the topic (P14, P19)—for example, to *"see the relationship"* (P19) between multiple topics discussed with the agent. When using just a single chat session, some participants (P9, P14, P19) would use verbal cues when switching topics (e.g., by saying *"let's change topics now"*).

Participants also used other strategies, including:

- Deleting or revising previous chat messages (P18).
- Using 'temporary chats' to prevent info saving (P4, P18).
- Changing the model in use (e.g., switching from GPT to Llama) (P11, P12, P14, P15).

- Repeating or emphasizing certain information or instructions in order to get the agent to remember these better (P4-P6, P9, P12, P14, P19).
- Asking the agent to summarize what it knows, in case the user is unsure (P16-P18).

However, while on the surface level many of these actions affect the storage and management of LTM (Stage 2), our participants expressed that they take these approaches with the goal of influencing when and how the agent retrieves memories and how it interprets them (Stage 3). For instance, most participants expressed that they try to abstract the agent's access to information in its LTM based on the current context of the conversation, such as the current task, project, or domain (e.g., separate memories related to writing from those related to health advice). Several participants (P1, P3-P6, P8, P15, P17-P19) even tried to keep memories from different projects, tasks, or domains disjointed from one another in the system, and when working on one task, they did not want the system to have access to memories from other unrelated tasks (e.g., they did not want the system to have access to memories about personal health advice when helping the user with academic writing).

3.2.3 Users' Challenges and Unsatisfied Needs with Agent Memory. Our participants' practices with memory management reflect their desires for agency over how and when the agent retrieves and interprets past memories in the present moment, and to have greater knowledge of what prior memories the agent is recalling and how it is using those memories in the moment. However, despite users' practices, many participants still faced challenges and expressed needs that were unsatisfied by current systems.

For instance, while most LLM agents interfaces allow users to view the contents of the user's chat histories, and some tools like ChatGPT allow users to scroll through a list of 'long-term memories' that are used across all chats, many of our participants who have used these features expressed that they are still unsure about when these memories are being retrieved and how they are being used and interpreted to influence the agent's responses. Participants also expressed a desire for greater control over how the agent retrieves and interprets memories, beyond their current strategies of organizing memories into chat threads or creating custom agents.

> *"So you know that that is like, you know, in the model's brain, there is a highlighter going over that word, that concept, whatever that information is. So, I would love to see what those are, and be able to manipulate them or edit them."* — P18

Agents sometimes misinterpret the intent of information in memory. For example, P17 described an instance when she was asking ChatGPT to refine a recipe, and asked it to *"make sure the salt is balanced."* ChatGPT interpreted this as a longer-term preference for all recipes moving forward, rather than just a preference for that particular recipe. While P17 was initially surprised and later amused by this, it reflects a challenge: agents can misinterpret users' prompts, and these misinterpretations can affect the agent's outputs and behaviors in undesired ways over time.

In some instances, the agent expresses that it remembers certain information, but it does not interpret or use that information in ways that the user wants it to. For example, P17 described that she uses ChatGPT for coding assistance on a large Unity project, and she keeps a single chat thread going for this project. She told ChatGPT that she is using Unity version 6, and the agent then expressed that it will remember moving forward. However, it still kept giving the user guidance that was relevant to previous versions of Unity rather than the newer version that she was using. This could have been due to the agent not knowing when that piece of knowledge in its LTM should be retrieved and how it should be used in the current context. In this case, P17 had to repeatedly remind the agent that she was using Unity version 6 and explain to the agent why that knowledge was important and how the agent should use that knowledge moving forward. Upon doing so, the agent began to make more proper use of that knowledge.

4 Study 2: Users' Comprehension of Agent's Interpretation of Discrete Memories

Through the first study, we found that participants' "memory management" efforts primarily target how the agent retrieves and interprets previously stored information at response time, whereas most deployed agent memory features primarily expose memory as lists of discrete stored items. This creates a concrete validation question: Does a discrete-memory-list representation support the user's goal of tracing an output back to the specific memory items that shaped it? In this second study, we therefore evaluate traceability between discrete memory items and the agent's output under a common memory-management representation. We used an interactive web-based system as an apparatus that exposes memory as a list of discrete items and allows users to include or exclude items when generating an agent response. Across multiple scenarios, we asked participants to identify which specific memories the agent used to produce a given output. We ask three research questions:

(1) Under a discrete-memory-list representation, can users correctly identify the specific set of memories the LLM used to produce a target response?
(2) Do users employ systematic strategies rather than random selection?
(3) How much does scenario type drive difficulty in memory matching and users' ability to converge?

4.1 Study 2 Method

Study 2 was conducted remotely: participants completed the task unsupervised, on their own devices and in their own environments, through a web application they accessed via Prolific.

Participants were presented with a *User Prompt* (scenario) along with a target *Desired Output* (the target assistant's answer), and they had access to a list of the agent's *memories* (17–21 short facts about the user) that could potentially be relevant. The task was to select the specific 3–5 memories that users believed were used by the AI to generate the Desired Output. Participants could toggle memories on or off and observe in real time how the AI's *Current Output* changed based on their selection. Importantly, this task does not evaluate whether participants personally preferred the Desired Output or whether it was the "best" answer. Instead, it evaluates traceability: whether a discrete-memory-list representation allows users to connect stored memory items to the agent's produced interpretation in an output. Fixing a Desired Output provides an

objective reference point for assessing whether participants can recover the underlying memory subset that generated it.

We recruited 100 participants from Prolific, restricting eligibility to adults fluent in English. After data cleaning (detailed below), we retained $N = 80$ participants for analysis. Each of the 8 main scenarios was attempted by roughly 14–18 different participants (total of 132 analyzed scenario trials; per-scenario counts: s1=18, s2=15, s3=16, s4=17, s5=16, s6=14, s7=18, s8=18). Ages ranged from 18–62 years (median = 32). The sample was 36.4% female, 59.7% male, and 3.9% undisclosed (self-reported). Countries of residence were primarily Europe (36.4%) and Africa (35.1%), followed by North America (18.2%), Asia (2.6%), South America (2.6%), and Oceania (1.3%). We did not collect a dedicated measure of prior LLM experience for Study 2; while experience could in principle moderate performance, the traceability goal we evaluate is intended to be achievable for novices as well as experienced users, and we did not observe obvious qualitative stratification in success during piloting.

The procedure was approved by the research ethics board, and all participants provided informed consent. The session begins with consent and a short welcome modal with instructions, followed by a practice scenario that teaches the mechanics. Each participant tackled two main scenarios (randomly assigned out of the 8, with scenario order randomized). We distributed scenario assignments such that each of the 8 scenarios was attempted by a comparable number of participants (ensuring counterbalancing across the sample). Following the study, participants were asked to answer a multiple-choice question: *"What did you consider when turning on/off memories to get to the desired output?"* For each scenario, participants followed the same procedure:

(1) Read the User Input and Desired Output;
(2) Select three to five memories that they believe drove the Desired Output;
(3) Inspect the Current Output generated from their selection;
(4) Iterate until satisfied, then proceed to the next scenario. Backtracking to the previous scenario is not allowed. A timer is set for seven minutes and displayed to the user with reminders at three minutes, two minutes, and one minute. The scenario automatically submits at timeout.

We developed a custom web application that simulates an AI agent with controllable memory (implementation details in Section G of the supplementary material). The interface (see Section C of the supplementary material) was divided into three panes: a list of memory items with checkboxes (the Memories pane), the AI's Current Output (which updates based on selected memories), and the static Desired Output for reference. Participants could select or deselect any memory by clicking the checkbox; each unique selection combination was sent to an LLM to generate a Current Output. The memory pane deliberately mirrored the discrete, list-based memory representations that current commercial assistants expose to users, though the application was purpose-built for this study rather than a replica of any particular product. Eligibility was not conditioned on which AI tools participants used regularly, and we did not administer a manipulation check on how realistic participants found the simulation. To ensure consistency, the system cached the AI's response for each unique combination of memory selections, so identical selections always yielded identical outputs. We used OpenAI's GPT-4.1-nano model with default parameters for generation (see Section D of the supplementary material for the prompt template); the rationale for this choice is given in Section G of the supplementary material.

We curated a total of 9 scenarios: 1 practice scenario and 8 main scenarios. Each scenario was designed to mimic a personal digital assistant use-case (e.g., fitness advice, travel planning, learning a language, daily routine optimization), and each came with a unique set of personal memories and a user query. These stimuli were chosen to mirror high-level topical groupings common in large public LLM conversation datasets (WildChat [49] and LMSYS-Chat [50]). The practice scenario (always presented first) was a simplified task with only 5 memories to choose from. The 8 main scenarios each provided 17–21 memory items and had a hidden ground-truth subset of 3–5 memories that were used to generate that scenario's Desired Output. See Section E of the supplementary material for the example of the scenario, memories, and desired output. We pre-generated each Desired Output by prompting the LLM with the user query plus the ground-truth memories, using a standard prompt template (identical to what would be used during the user's interactions). By designing scenarios across a variety of domains, we aimed to cover both easier cases (where relevant memories are more obvious) and harder cases (where the relevance is more subtle), allowing us to examine scenario-driven difficulty (RQ3).

We applied a rigorous data cleaning procedure (pre-registered prior to data collection) to ensure that only engaged interactions were analyzed. First, any participants who failed to follow instructions or complete the study properly were removed. In total, 7 participants were excluded because they attempted to circumvent the interface flow (for example, by refreshing the page to restart or skip scenarios, or by quitting early). Next, we excluded individual scenario trials that did not meet minimum engagement criteria. If a participant finished a scenario in under 60 seconds or toggled fewer than 3 memories in total, we considered that trial invalid (since our instructions explicitly told users to take time exploring different memories). This led to the removal of 54 scenario-level records. We also excluded the practice scenario from all analyses.

The system automatically logged all participant interactions with timestamps. Every memory selection or deselection event was recorded along with the current set of selected memory IDs and the corresponding output text generated by the model. This rich log allowed us to reconstruct each participant's trajectory through the memory-selection space over time. For each action (each toggle of a memory), we compute the overlap between the participant's selected set S_t and the scenario's ground-truth memory set G. Our primary accuracy measure is the Jaccard similarity $J(S_t, G) = \frac{|S_t \cap G|}{|S_t \cup G|}$, which ranges from 0 (no overlap with the correct set) to 1 (exact match of the correct set). This measures how close the participant's selection is to the true solution at any given moment. For RQ1, we examine overall user performance in identifying correct memories by looking at max Jaccard score and whether participants eventually found a majority of the correct set. For RQ2, we analyze the interaction logs to see if participants improved their selections systematically over time. We compute each participant's Jaccard trajectory metrics (e.g., monotonicity – proportion of non-decreasing steps) and summarize common patterns. For RQ3, we

investigate scenario-level differences. We treat scenario (content type) as a factor and analyze its effect on task success and difficulty. Because each participant saw only two scenarios, scenario differences are essentially between-subject. We used a mixed-effects modeling approach to account for variability due to participant and scenario. Specifically, we fit linear mixed models for key metrics (e.g., final Jaccard, number of toggles) with scenario as a fixed effect and a random intercept for each participant (to account for individual differences in overall skill). This allowed us to test if certain scenarios were significantly harder (lower accuracy or required more toggles) than others. All statistical tests use $\alpha = 0.05$.

4.2 Study 2 Results

4.2.1 Memory Identification Performance (RQ1). Participants were largely unable to recover the specific memories that the AI had used to produce its answers. On average, the final Jaccard similarity was only 0.181 (and a maximum of 0.314). Many users stumbled upon a better partial solution at some point but then moved away from it by the end, failing to lock in their highest-scoring memory combination. Complete success was exceedingly rare: only around 18 % of the trials ended with the user even identifying a majority (60 %) of the ground-truth memory set, and almost no one achieved a perfect 100 % match. Participants struggled to infer which pieces of stored information the agent actually used and how, underscoring a misalignment between how current memory systems function and how users understand the agent's memory in context.

4.2.2 User Strategies and Trajectories (RQ2). Despite the low overall accuracy, participants did not simply toggle memories at random. On average, users performed roughly 19.7 selection or de-selection actions per scenario and spent about 3–4 minutes actively refining their choices, indicating a serious, systematic exploration of the memory space. Most participants' performance improved over time: the mean trajectory monotonicity was around 0.663 (on a 0–1 scale), meaning that about two-thirds of memory-selection changes either maintained or increased the Jaccard similarity relative to the previous state. In other words, users tended to move closer to the correct set as they iterated (answering RQ2 in the affirmative), albeit gradually and not without setbacks.

However, this improvement was far from a smooth, linear climb. Participants' Jaccard trajectories often plateaued or even dipped before rising again. It was common for a participant to reach a Jaccard plateau (often around 0.2–0.3) and struggle to push beyond it by toggling additional memories on or off. Such patterns suggest that participants were actively reasoning about which memories might matter—experimenting with various plausible subsets–yet lacked clear feedback on *why* a given selection was or was not working. Self-report data from the post-task survey further confirm that participants employed deliberate strategies rather than random guessing. For example, 74 of 80 respondents cited "relevance to the main topic" and 68 cited "specific details mentioned" as strategies. By contrast, far fewer participants reported using cues like the memory item's emotional tone, its chronological order, or its personal importance to the user. This emphasis on topical and detail matching suggests that users attempted to logically deduce which facts the AI would need in order to produce the given answer. However, even these sensible strategies often fell short. Participants could identify obviously relevant facts, yet still miss the subtler cues that the AI actually used to choose its words, or misinterpret which details were truly influencing the output.

4.2.3 Scenario-Dependent Difficulty (RQ3). Memory-matching difficulty varied across scenarios (see Section F of the supplementary material). Scenarios s6 and s8 were hardest (mean max Jaccard only 0.163 and 0.296, respectively) while s7 and s2 were easiest (0.421 and 0.378). One more example: while 30% of trials in s7 reached majority overlap, none did so in s6 (see Section F of the supplementary material). This pattern suggests that some scenarios (e.g., those with fewer distractors or clearer cues) were easier for memory matching, whereas others (like s6) were challenging. We modelled final Jaccard similarity with a linear mixed-effects model including scenario as a fixed effect and a by-participant random intercept, fitted by maximum likelihood to the 132 analysed scenario trials (residual variance 0.0168; by-participant intercept variance 0.003). Taking s1 as the reference ($\beta = 0.165$, $SE = 0.033$, $z = 4.99$, $p < .001$), two scenarios differed reliably from it: s2 ($\beta = 0.105$, $SE = 0.046$, $z = 2.29$, $p = .022$) and s7 ($\beta = 0.119$, $SE = 0.047$, $z = 2.56$, $p = .011$). The remaining contrasts were not reliable: s3 ($\beta = -0.070$, $SE = 0.048$, $z = -1.45$, $p = .146$), s4 ($\beta = 0.072$, $SE = 0.047$, $z = 1.53$, $p = .125$), s5 ($\beta = 0.021$, $SE = 0.048$, $z = 0.45$, $p = .657$), s6 ($\beta = -0.077$, $SE = 0.050$, $z = -1.54$, $p = .124$), and s8 ($\beta = -0.048$, $SE = 0.047$, $z = -1.02$, $p = .307$). Inference is based on Wald z tests, which do not define residual degrees of freedom.

5 Discussion

Overall, our findings shed light on users' perceptions, mental models, and practices with agent memory as they relate to the different stages of memory flow in agents (intake, storage and management, retrieval, and interpretation, as illustrated in Figure 1). Our findings reveal that users' perceptions and mental models of agent memory are mainly influenced by agents' behaviors, which are affected by how agents *retrieve* and *interpret* (Figure 1, Stage 3) information from LTM. As a result, users' practices with managing and exploring agent memory are mainly an attempt to influence and understand how the agent retrieves and interprets information from LTM, rather than strictly attempting to affect information storage and management on its own. However, when memory is exposed as a bounded list of discrete items, users often cannot reliably trace an agent's output back to the specific memories that shaped it, suggesting that discrete-list controls may be structurally misaligned with users' Stage-3-oriented goals.

In addition to communicating to users what information is being stored in the agent's LTM, users can benefit from the agent communicating with the user about how it is retrieving and interpreting memories. One possible approach for this is a conversational approach, where it states explicitly what it is recalling from memory and describes how it is using or reasoning with that memory to generate a response to the user's current query. It can also provide implicit conversational cues—for example, by regularly bringing up facts that it has learned about the over time in conversation (e.g., saying things like *"given that you are an engineer, you might prefer to..."* or *"this recipe might be a good way to satisfy your preference for oranges*", where the underlined statements are 'facts' contained in the agent's LTM that the agent brings up to assure the user that

it is regularly thinking about these memories). Another possible approach is a GUI-based approach—for example, where UI indicators next to the chat interface show which information from LTM is retrieved and how it is used to generate the agent's responses.

Approaches for user interaction with agent memory should also be designed to give users more influence on how information from the agent's LTM is retrieved and interpreted. Our findings highlight that how memories are surfaced from the agent's LTM should be based on the user's intentions. Therefore, we recommend that agents should infer the user's intention (e.g., by inferring the user's current task or context), or *ask* the user about their intention. For instance, the agent can ask the user if they think it is relevant to retrieve memories about a certain topic before doing so—e.g., if the user is conversing with the agent about trip planning, and the agent thinks that their previous conversation about exercise planning might be relevant to the current trip planning task (e.g., planning trip activities around the workout plan), then the agent could ask the user before retrieving or using those memories.

Users could also benefit from controlling access to memories by task. Beyond existing approaches that let users manipulate memories directly—moving them between conversations or editing their level of abstraction [18, 45]—organizing memories into hierarchical structures (akin to folders in file systems, or scoping in object-oriented programming), where users define the appropriate level of access per task, could align well with our participants' expressed needs.

Our Study 2 findings reveal low traceability (i.e., scarce, trustworthy signals) about how personal information shaped a given LLM output. If users cannot reliably infer which pieces of their data were invoked or how that data was interpreted, then meaningful consent and accountability become fragile: people may agree to storage but still be unable to anticipate downstream use at the point of retrieval and interpretation. Recent interview work on RAG-based memory similarly documents incomplete mental models and a desire for granular control and transparency over how memories and inferences are used [46]. This aligns with public concerns raised when consumer systems introduce opaque "memory" features and default retention policies [1, 11], and with life-logging scholarship that ties empowerment to clear boundaries and user control over capture and reuse [31]. Complementarily, design probes that surface and visualize sensitive content in memories (e.g., MemoAnalyzer) illustrate how retrieval-time auditability can raise awareness and enable correction [38, 47]. Our results, therefore, motivate attention to defaults, disclosure, and revocation as retrieval-time phenomena as much as storage-time settings, and invite community dialogue on what forms of user-facing audit (seeing what was remembered and why it was used), what scopes and expirations, and what propagation of revocation to summaries and learned associations make interpretation practices legible without overburdening users.

Participants often reasoned with human-like analogies (episodic cues, context-triggered recall), yet the systems they used can retain and search information in non-human ways. This reframes a familiar debate: whether LTM for assistants should lean toward human-compatible behaviors or embrace AI-native capabilities. Historical evidence suggests that human-compatible behaviors may aid comprehensibility and comfort [19, 25], while AI-native behaviors promise power at the risk of opacity. A productive near-term path is to explore bridges between the two: for example, internal mechanisms that exploit AI strengths paired with user-facing representations that remain legible and choiceful [30].

The insights from this work are based on users' recollections from their everyday usage of agents and on behavior within a controlled memory-matching task that bounded candidate memories and limited interaction time. A further trade-off is that the scenarios used a fixed Desired Output rather than allowing participants to pursue an open-ended response they personally wanted. This choice is necessary for a controlled test of traceability: without a shared target output, there is no objective basis for evaluating whether participants can recover the memory subset that produced a particular interpretation. In real deployments, users often care about both preference and provenance; our goal here was to isolate the provenance question. These choices strengthen internal validity around the locus of difficulty we observe at retrieval and interpretation, but they also limit ecological validity: real deployments often involve larger, noisier, and sometimes multimodal memory stores. A further consideration is that participants reasoned about a fictional user's memories rather than their own. Because they had no first-hand knowledge of the person those memories described, they could not draw on the personal context that ordinarily makes a memory feel relevant, and the low traceability we observe may therefore partly reflect unfamiliarity with third-party data rather than an inherent limit of discrete-memory representations. Whether users can better trace an agent's interpretation of memories accumulated through their own interactions remains an open question, and one we see as a priority for future work. For Study 1 we did not systematically collect participants' occupations, educational backgrounds, or countries of residence, which limits what can be said about the transferability of the interview findings. Moreover, real systems may exhibit substantial non-identifiability, where multiple memory subsets can produce similarly acceptable outputs; if anything, this would make traceability harder in the wild and strengthen the case for retrieval-time audit and explanation mechanisms beyond discrete memory lists. More insights can be gained by developing, exploring, and comparing different approaches to user interaction with LTM; therefore, we recommend this as an avenue for future research. We also recommend longitudinal studies to gain insights into users' perceptions and interactions over time, as well as studies that examine multiple users' interactions with memory in shared agents (rather than just agents "owned" by individual users).

6 Conclusion

We identified a worrisome misalignment between users' Stage 3 goals (steering and auditing retrieval-time interpretation) and common Stage 2 "discrete data" memory-management approaches; in particular, Study 2 evidences a traceability/provenance gap in which users cannot reliably map outputs back to the specific memories that shaped them. Therefore, there are opportunities for technological approaches to user interaction with agent LTM to focus on providing control and understanding of memory retrieval and interpretation.

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Supplementary Material for "What Does It Mean To Remember? Understanding Users' Perceptions and Practices with Long-Term Memory in AI Agents"

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A Study 1 Semi-Structured Interview Guide

Interviews were semi-structured, and the exact questions asked occasionally deviated slightly from the following guide.

A.1 Introduction

(1) Share with me how you currently use personalized AI tools like ChatGPT or Claude.
   - **What AI tools** or chatbots do you use on a regular basis?
   - What do you use these tools for? (What types of **tasks/activities**?)

(2) Share with me (in general) the **role that memory plays** in your use of these systems.
   - What things does the system remember over time?
   - How does the system's memory affect the task(s) it does (or helps you do)?

A.2 Users' Mental Models of and Desires from Agent Memory

(3) Share with me your understanding of how the memory in [X system] works.
   - How does it **store** memory (how does it remember things)?
   - How does it choose **what to remember**?
   - How does it **recall** memories?
   - How does it **use** memories?
   - How does it **behave** based on its memories? How do you **expect it to behave** based on its memories?

(4) What kinds of behaviours from the system (e.g., indications, communications, interface elements, etc.) **help you better understand the system's memory operations**?
   - Is there anything that [X system] does that you think helps you better understand how the system's memory is working (e.g., what it is remembering, recalling, forgetting, etc.)?

A.3 Users' Practices with Agent Memory

(5) During your normal interactions with the agent, is there anything that you do with it specifically to **try to get it to remember certain things** (or forget certain things)?
   - Is there anything you do with the system to **try to *prevent* it from remembering certain things**?

(6) In what instances do you **want to be more aware** of what is in the system's memory?
   - In what instances do you want to know what the system is choosing to remember? (In what instances do you want to control this?)
   - In what instances do you want to know what and how the system is recalling information in its memory? (In what instances do you want to control this?)
   - In what instances do you want to know what the system is choosing to forget? (In what instances do you want to control this?)

(7) In what instances do you want to **manage the system's memory**?

A.4 Conclusion

(8) Is there anything else that you would like to share?

B Overview of Study 1 Participants

PID	Age	Gender	Experience Using AI	AI Agents/Tools Used	Summary of Tasks Used For	Expertise Using AI (/5)	Technical Knowledge of LLMs (/5)
P1	30	Female	At least once per week	ChatGPT	Writing assistance, idea generation, language translation, audio transcription, coding assistance, searching information	4	4
P2	35	Female	At least once per day	ChatGPT, Claude, Copilot	Personal tutor, searching information, companionship	4	2
P3	27	Female	At least once per day	ChatGPT, Grammarly AI tool	Productivity tasks, writing assistance, usability analysis	4	4
P4	24	Male	At least once per week	ChatGPT, Claude, Perplexity	Coding assistance, brainstorming, problem solving, writing asssistance, learning, math problems, searching information	4	4
P5	25	Male	At least once per day	ChatGPT	Brainstorming, writing assistance, data analysis	4	1
P6	33	Female	At least once per week	ChatGPT, Claude, Copilot	Writing assistance, coding assistance	4	4
P7	26	Male	At least once per day	ChatGPT, Grok	Studies, research	5	5
P8	28	Male	At least once per day	ChatGPT, Grok, Replika	Studies	5	5
P9	35	Male	At least once per day	Replika, Kuki AI (Mitsuku), SimSimi	Conversations, emotional support, entertainment, language learning	5	5
P10	24	Male	At least once per day	ChatGPT, NotebookLM	Studies	4	4
P11	—	Male	At least once per day	ChatGPT, DeepSeek, Claude	Coding assistance, idea generation, planning, writing assistance, studies	5	5
P12	25	Female	At least once per day	Replika, ChatGPT	Companionship, conversations	5	5
P13	28	Male	At least once per day	ChatGPT, NotebookLM	Searching information, planning	5	4
P14	25	Female	At least once per day	Replika, ChatGPT	Brainstorming, idea generation, journaling, reflection, emotional support, conversations, personalized recommendations, writing assistance, summarizing documents	5	5
P15	60s	Female	At least once per day	Claude, ChatGPT, Cursor (AI code editor), Warp (AI terminal)	Foreign language learning, language translation, coding assistance, writing assistance, searching information, question answering, photo interpretation, brainstorming, idea generation, companionship, conversations, computer terminal usage	5	3
P16	24	Male	At least once per day	ChatGPT	Research, brainstorming, idea generation, planning, writing assistance	4	3
P17	28	Female	At least once per day	ChatGPT	Cooking, recipes, finance (tax questions), searching information, coding assistance, academic research	3	3
P18	30	Female	At least once per day	ChatGPT, Custom GPTs, NotebookLM	Analytical thinking, creative thinking, coding assistance, reflection	4	4
P19	28	Female	At least once per day	ChatGPT, NotebookLM	Studies, research, conversations, companionship, summarizing documents	5	3
P20	30	Male	At least once per day	ChatGPT	Studies, research, idea generation	5	5
P21	21	Male	At least once per day	ChatGPT	Studies, research	5	3

C Study 2 Interface

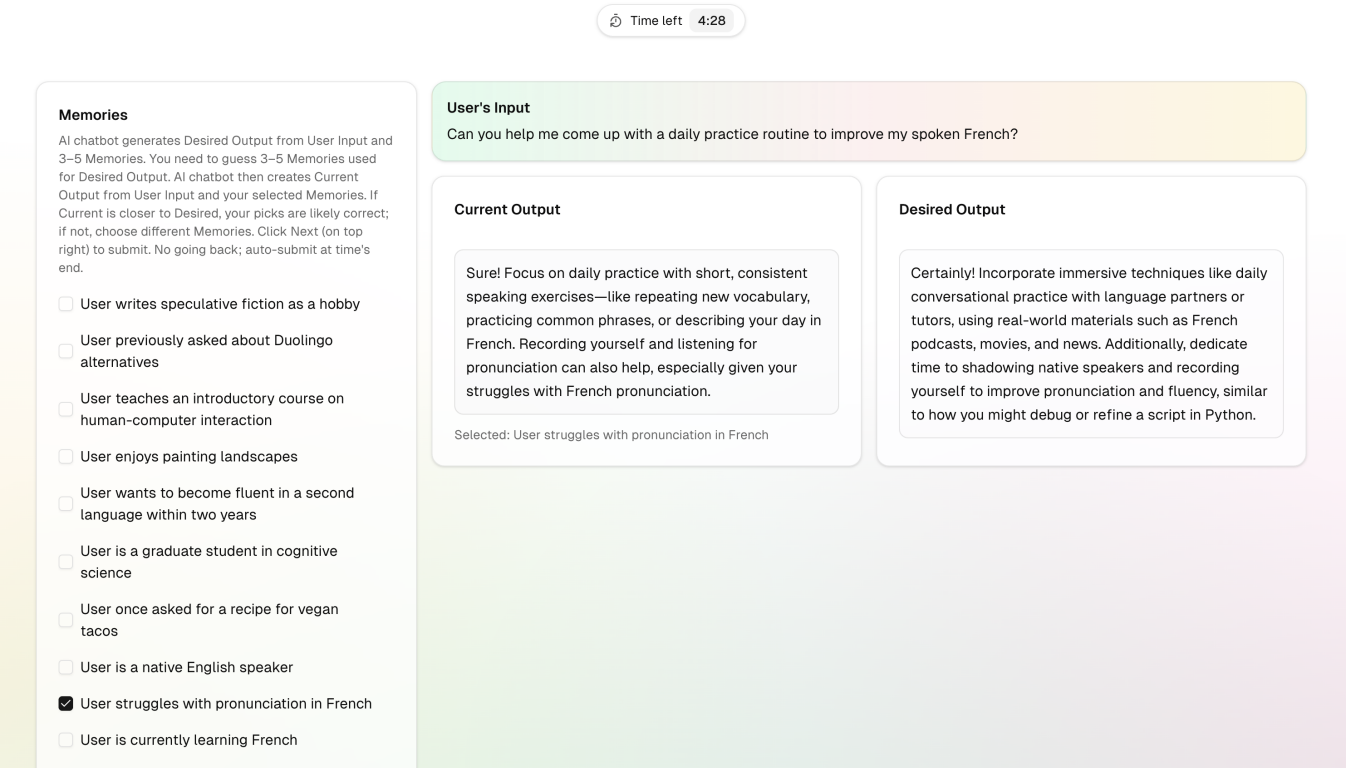


Figure 1: Interface used in Study 2. Participants view a countdown timer, read the *User's Input* prompt, and select *Memories* from a checklist (left). The system displays a generated *Current Output* (center-left) alongside the *Desired Output* target (center-right).

D Study 2 LLM Prompt Template

For Study 2, we used a consistent prompt template to query the language model. The assistant was given a system message providing context about its long-term memory and the user's query, followed by the user prompt. The exact system message template is shown below (with a placeholder for the memory list):

```
You are an intelligent agent with long-term memory. You are to respond
to the user's prompts intelligently, using the injected long-term
memories as relevant context when responding.

The following is a list of the injected long-term memories, to use
as relevant context:

```
{
  "memories": [
    "Memory text 1",
    "Memory text 2",
    "... etc. ..."
  ]
}
```

When responding to the user, make your response short (at most one
paragraph of 2-3 sentences), and consider all of the above long-term
memories as much as possible.
```

In this template, the JSON-formatted list of memory items (relevant personal facts) is injected into the system message. The user's actual query is then provided as the user message. This prompt ensured that the model considered the supplied memories when generating its response.

E Study 2 Example Scenario, Memories, and Desired Output

To illustrate the task in Study 2, we provide an example scenario along with its memory items and the AI's Desired Output. The following is the practice scenario (used as a tutorial for participants):

User Prompt (Scenario): "I've been feeling more fatigued during my morning runs lately. Can you suggest some adjustments to help boost my endurance?"

Memory Items (shown to the user):

- M1: User lives in a coastal city with mild winters. *(Ground-truth relevant memory)*
- M2: User is pursuing an online certification in digital marketing.
- M3: User runs 5 kilometers three times a week. *(Ground-truth relevant memory)*
- M4: User avoids caffeine after 2 PM to improve sleep quality. *(Ground-truth relevant memory)*
- M5: User is learning to play the electric guitar.

(*In this scenario, the true set of memories used by the AI comprised M1, M3, and M4.*)

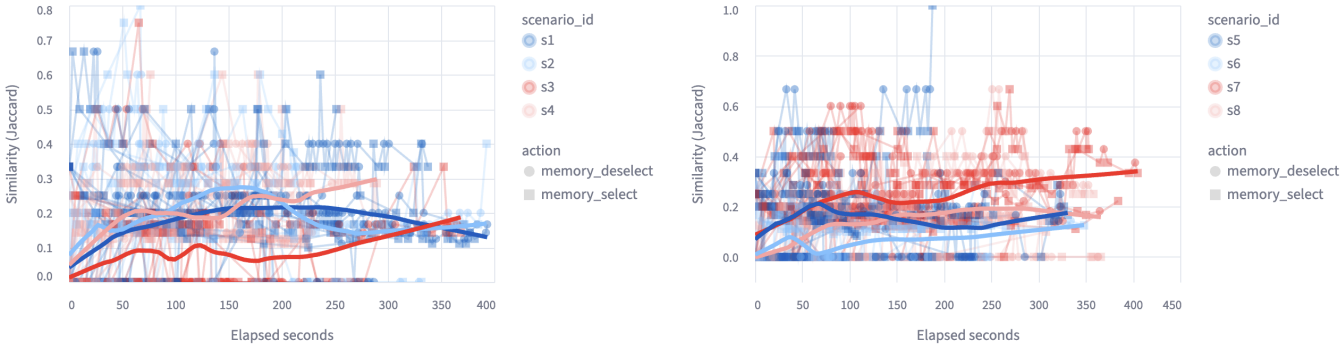
AI's Desired Output: "Since you run 5k several times a week, one way to boost your endurance is to gradually increase your distance or pace over time. Make sure you're getting good rest—it's great that you avoid late-day caffeine to help with sleep, as quality sleep improves endurance. Living in a place with mild winters also means you can train consistently year-round, so taking advantage of that with a regular routine and maybe some cross-training will help build your stamina."

This Desired Output was generated by the assistant using only the ground-truth memories (M1, M3, M4) as context. Participants in the study saw the User Prompt and the list of memory items, and their task was to select which 3–5 memories they believed the assistant used to produce the Desired Output. The example above demonstrates how the relevant memories (marked in *italic*) inform the content of the AI's response.

Table 1: Per-scenario metrics. Max Jaccard is the mean peak similarity, and "Majority" is the fraction of trials attaining $\geq 60\%$ of the true memory items.

Scenario	Avg. Max Jaccard	% Trials Majority
s1	0.352	22.6%
s2	0.378	35.7%
s3	0.206	11.8%
s4	0.362	14.7%
s5	0.299	25.8%
s6	0.163	0.0%
s7	0.421	30.3%
s8	0.296	6.9%

F Study 2 Per-scenario metrics



(a) Scenarios s1–s4.

(b) Scenarios s5–s8.

Figure 2: Similarity trajectories across all study scenarios. Each shows four colored trend lines with many faint action points (circles for memory deselect, squares for memory select). Left panel covers s1–s4, right panel covers s5–s8. The Y-axis is the Jaccard similarity between the Current and Desired outputs; the X-axis is elapsed seconds.

G Study 2 Implementation Details

The study application was built with Next.js/React and deployed on Vercel with a Supabase/Postgres backend.

The system cached the AI's response for each unique combination of memory selections encountered for a scenario, and reused cached outputs whenever the same combination reoccurred. This caching meant that identical memory selections always yielded identical outputs, and previously seen outputs could be retrieved instantly, allowing participants to reliably compare the effect of toggling a particular memory.

We used OpenAI's GPT-4.1-nano with default parameters. We deliberately chose this model over newer but slower alternatives because the latter introduced significant queue delays (4–6 seconds before processing) that would have undermined the real-time interactivity of the task.

Each of the 8 main scenarios provided 17–21 memory items. We chose this range to balance task complexity with experimental feasibility: the memory set needed to be large enough to require meaningful discrimination among relevant and irrelevant memories, while remaining small enough for participants to complete each scenario within the target time established during pilot testing.